

Petr Boska Nylander

Professional Summary

- MSc in Machine Learning from DTU with two years of teaching assistant experience and a thesis on predicting water pipe failures for Danish utilities.
- Experience building machine learning projects from raw data to modelling, evaluation, and reproducible Python pipelines, both independently and in teams.

Education

Jan 2024-Aug 2026 **MSc Mathematical Modelling and Computation**, *Technical University of Denmark*.
Specializing in Machine Learning and Data Science.
MSc Thesis: Predicting water pipe failures with survival analysis and machine learning, combining pipe records from three of the largest Danish utilities with soil, climate and geospatial data
Courses: Deep Learning, Computer Vision, MLOps, Time Series Analysis, Bayesian Machine Learning, Optimization, Multivariate Statistics, High-Performance Computing
Notable Projects: Synthetic medical image generation with diffusion models, MLOps CAPTCHA pipeline, large-scale graph mining on the Yelp dataset, LLM reasoning robustness, deep generative models, medical image segmentation, object detection, image classification

Aug 2020-Jan 2024 **International Bachelor in Natural Science**, *Roskilde University*.
Specialized in Mathematical Modelling.

Aug 2023-Jan 2024 **Machine Learning (Guest Course)**, *IT University of Copenhagen*.

Summer 2021-2022 **Additional Courses**.
Deep Learning (University of Southern Denmark), Data Structures & Algorithms (IT University of Copenhagen)

Work Experience

Aug 2023-Dec 2023 **Teaching Assistant**, *Roskilde University*, Calculus.
Guided Natural Science undergraduates through differential and integral calculus concepts, conducted exercise sessions, and provided assignment feedback.
Contact Person: Johanne Gudmand-Høyer (joguho@ruc.dk)

Aug 2022-Dec 2023 **Teaching Assistant**, *Roskilde University*, Introduction to JAVA.
Taught computational thinking, algorithms, and OOP to undergraduates. Conducted exercise sessions and evaluated student assignments.
Contact Person: Jens Classen (classen@ruc.dk)

Aug 2022-Oct 2022 **Teaching Assistant**, *Roskilde University*, Statistics and Data Analysis.
Assisted undergraduates with Python-based statistical analysis and visualization during exercise sessions and mini-projects.
Contact Person: Ulf Rørbæk Pedersen (urp@ruc.dk)

IT Skills

Languages Python, SQL

Tools PyTorch, scikit-learn, statsmodels, pandas, NumPy, GeoPandas, Docker, MLflow, Hydra, Git, Bash, LaTeX, Linux

Languages

Danish, English, Czech

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